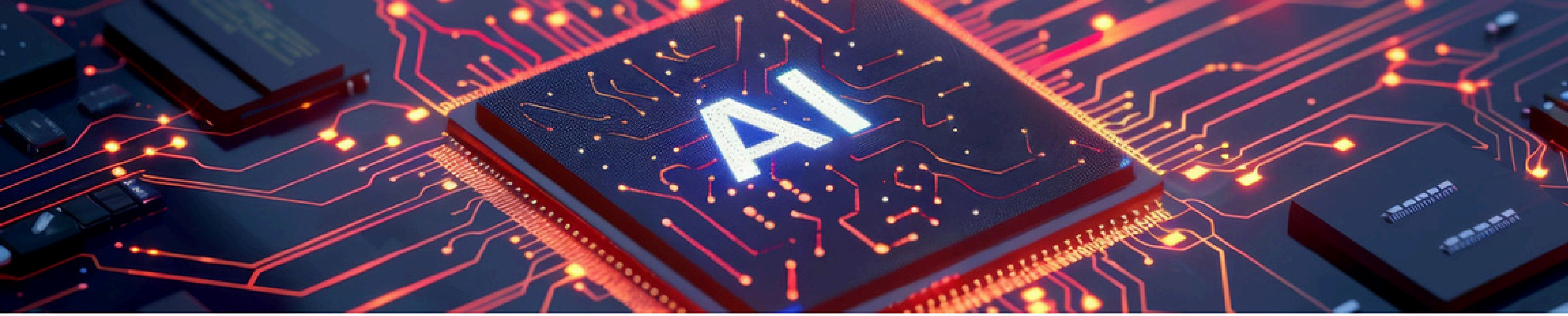

ML 24/25-01 Investigate Image Reconstruction By using Classifier

This presentation discusses innovative techniques for efficient image storage and restoration, leveraging KNN and HTM classifiers for enhanced performance.

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Presentation Agenda Overview

Detailed breakdown of presentation topics and flow

- Introduction & Motivation
  - Challenges of large-scale image storage
  - Need for efficient high quality restoration.
- Project Overview & Objectives
  - Hybrid framework using KNN and HTM
  - Goals: Reduce storage, maintain quality, and enhance reconstruction
- Methodology
  - Data Acquisition & Preprocessing
  - Feature Extraction via Spatial Pooling
  - Classifier Training (KNN & HTM)
  - Image Reconstruction & Fusion
- Evaluation & Results
  - Quantitative metrics and visualizations (tables, box plot, histogram)
  - Comparison: individual vs. combined performance
- Conclusion & Future Work
  - Summary of findings
  - Final thoughts and potential directions for future research.

Data Explosion and Storage Challenges

- **Digital Age Data Explosion**

  The digital age has led to an unprecedented increase in data generation, posing significant storage challenges.

- **Need for Efficient Storage**

  There is a growing need for efficient image storage solutions that maintain high-quality restoration capabilities.

- **Smart Patterns Concept**

  Inspired by brain functionalities, breaking images into 'smart patterns' can enhance storage efficiency and retrieval speed.

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Hybrid Image Reconstruction Framework

- **Hybrid Framework Overview**

  A hybrid image reconstruction framework integrates KNN and HTM classifiers for enhanced performance.

- **Objectives of the Project**

  The project aims to reduce storage needs while maintaining image quality during reconstruction.

- **Storage Efficiency**

  Focus on minimizing storage requirements without compromising the quality of the reconstructed images.

- **Fusion of Classifiers**

  Utilizes a fusion of KNN and HTM classifiers to improve accuracy in image reconstruction tasks.

- **SDR Generation**

  Employs the NeoCortex API to generate Sparse Distributed Representations (SDRs) for enhanced analysis.

Data Acquisition & Preprocessing

Understanding Fashion MNIST dataset processing

- **Dataset Overview**

  The Fashion MNIST dataset includes 10,000 training images and 2,000 testing images for machine learning tasks.

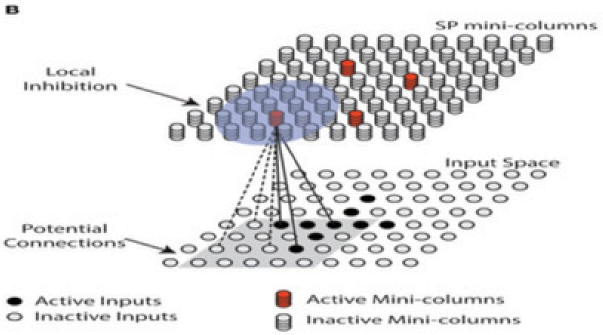
- **Binarization Process**

  This step converts PNG images into binarized text files, simplifying data handling. Using ImageBinarizer Nuget Package.

- **Vectorization Explained**

  Vectorization involves flattening binary data into a one-dimensional array for model input.

Feature Extraction via Spatial Pooling



- **Purpose of Spatial Pooling**

  Spatial pooling extracts high-dimensional feature representations called SDRs, essential for image analysis.

- **Utilization of NeoCortex API**

  The NeoCortex API facilitates the feature extraction process, enhancing efficiency in capturing image features.

- **Robustness to Noise**

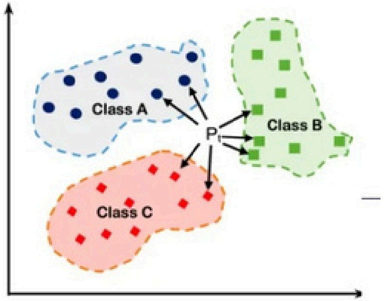
  SDRs provide robustness against noise, ensuring accurate feature representation despite image imperfections.

- **Capturing Essential Features**

  SDRs effectively capture crucial image features like edges and textures, vital for computer vision tasks.

AI-Enhanced: Craft a concise headline for skimmers

K Nearest Neighbors



- **KNN Classifier Functionality**
  KNN stores training SDRs along with their corresponding labels.

- **Classification Methodology**
  It employs weighted voting to classify test images based on overlap.

- **HTM Classifier Overview**
  HTM retains both SDR and original image vectors for classification.

- **Image Reconstruction with HTM**
  HTM reconstructs images using a weighted average of training examples.

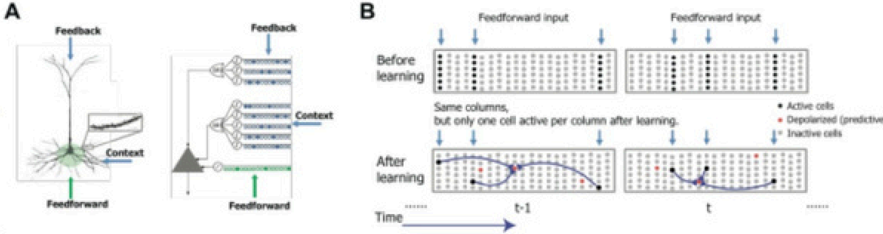


Image Reconstruction and Fusion Techniques

Exploring techniques in image reconstruction and fusion

- **Individual Reconstruction Techniques**

  Utilize Hierarchical Temporal Memory (HTM) and K-Nearest Neighbors (KNN) for effective image reconstruction.

- **Fusion Strategy Overview**

  Combine outputs based on cosine similarity scores to enhance reconstruction quality.

- **Cosine Similarity Application**

  Employ cosine similarity to evaluate and rank the outputs from individual classifiers for fusion.

- **Post-Processing Methods**

  Implement Gaussian weighted local voting and median filtering for refined image output.

- **Strengths of Classifiers**

  The fusion approach effectively leverages the strengths of HTM and KNN classifiers for better results.

Essential Evaluation Metrics for Image Quality

- **Cosine Similarity**
  Measures the similarity between original and reconstructed images through their vector representations.
- **Binary Similarity**
  Evaluates images by comparing each pixel to assess reconstruction fidelity.
- **Importance of Cosine Similarity**
  Highlights high-level structural similarity, crucial for understanding overall image quality.
- **Importance of Binary Similarity**
  Focuses on pixel-level accuracy, ensuring no detail is lost in the reconstruction process.

Results Summary of Classifiers

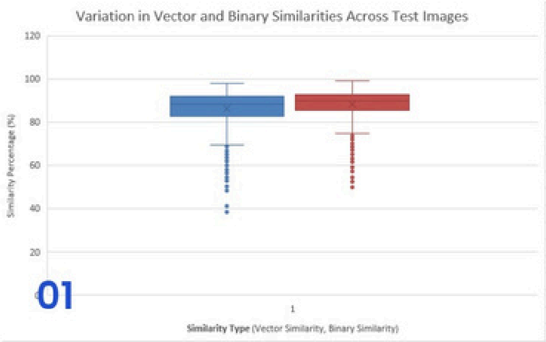
Comparison of classifier performance metrics

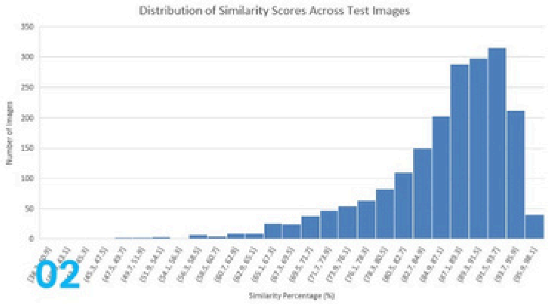
Classifier	Vector (%)	Binary (%)
KNN	83.39	87.00
HTM	85.90	87.28
Combined	86.22	88.12

Fusion Classifier Performance Analysis

Box Plot Overview

The box plot illustrates the distribution of similarity scores, highlighting median, quartiles, and outliers.





Histogram Insights

The histogram reveals the frequency distribution, showing most similarity scores clustered between 80–100%.

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Discussion & Model Comparison Insights

Exploring the Benefits of Fusion in AI Models

- **Fusion Approach Yields Highest Accuracy**
  The fusion method outperforms others, achieving the best accuracy in reconstruction tasks.
- **HTM and KNN's Strengths**
  HTM is strong in high-level features, while KNN enhances pixel-level accuracy.
- **Leveraging Strengths for Robustness**
  By combining HTM and KNN, the fusion approach ensures robust reconstruction outcomes.
- **Implications for Storage Efficiency**
  Higher accuracy methods can lead to improved storage efficiency in data management.

Conclusion and Future Work Summary

Recap of findings and future enhancements

- **Project Objectives Recap**

  The project aimed to analyze a hybrid approach's effectiveness in ML.

- **Key Findings Highlighted**

  Results confirmed that the hybrid model improved accuracy over standard methods.

- **Advantages of Hybrid Approach**

  Combines strengths of multiple techniques, enhancing overall performance.

- **Future Enhancements Suggested**

  Testing with larger datasets can improve model robustness and reliability.

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Thank You for Your Attention Team Code Avengers

Thank you for your engagement *For any inquiries or further information, feel free to reach out using the contact information provided.*

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